

PRATEEK SINGH

AI/ML Engineer & Data Analyst

 Milford, MA |  +1 (781) 363-8616 |  prateek_singh@waters.com |  LinkedIn |  GitHub

SUMMARY

AI/ML Engineer and Data Analyst with 7+ years of experience spanning enterprise AI development, data analytics, automation engineering, and system integration across global markets (USA, South Africa, India). Currently building production-grade Generative AI search platforms and enterprise automation systems at Waters Corporation. Proven track record delivering measurable impact — including 80-85% reduction in manual SAP operations and architecting RAG-based AI search across 1M+ documents. Deep expertise in Python, AWS cloud-native architecture, LLM orchestration, database engineering, and end-to-end data pipelines. Adept at cross-functional collaboration, client-facing delivery, and translating complex business requirements into scalable technical solutions.

TECHNICAL SKILLS

AI/ML & GenAI	LangChain, RAG Architecture, Prompt Engineering, Claude LLMs (Anthropic), Amazon Bedrock, Amazon Titan Embeddings, OpenSearch Vector DB, Vector Embeddings, RAGAS Evaluation, Synthetic Data Generation, Content Guardrails
Languages	Python, SQL (T-SQL, PL/pgSQL), R, Groovy
Cloud & Infra	AWS (Lambda, S3, Bedrock, API Gateway, CloudWatch, SageMaker, EC2, IAM), Serverless Architecture, Boto3 (AWS SDK), Azure DevOps, CI/CD Pipelines
Data & Databases	MongoDB, MySQL, PostgreSQL, SQL Server, OpenSearch, Data Modeling, Data Warehousing, ETL/ELT Pipelines
Frameworks & Tools	Streamlit, Pandas, NumPy, Scikit-learn, TensorFlow, Matplotlib, BeautifulSoup, WebSocket APIs, REST APIs
SAP Technologies	SAP Backoffice Commerce, HAC, ImpEx, FlexibleSearch
Visualization	Tableau, Power BI
Methodologies	Agile/Scrum, Git, GitHub, JIRA, ServiceNow

PROFESSIONAL EXPERIENCE

AI/ML Engineer

Nov 2024 – Present

Waters Corporation

Milford, MA

AI Search & Generative AI Platform

- Developed backend Python infrastructure for **AI-powered search summarization**, enabling Gemini-like generative responses on Waters website for product discovery across **1M+ indexed documents**
- Engineered **AWS Lambda functions** for serverless LLM orchestration using **Amazon Bedrock** and **Claude models**, implementing real-time streaming via **WebSocket APIs**
- Architected **RAG pipeline** integrating **OpenSearch vector database** with **LangChain** to index and retrieve product catalogs, application notes, and support documentation
- Implemented **content guardrails** ensuring domain-relevant AI responses, preventing hallucinations and maintaining brand integrity
- Developed **RAGAS evaluation framework** and **synthetic test data generation** measuring faithfulness, relevance, and context precision
- Created automated **web crawling pipeline** for continuous ingestion of website content into vector embeddings for semantic search

SAP Backoffice Automation Platform

- Architected **end-to-end automation platform** for SAP Backoffice Commerce, reducing manual effort from **3-5 days to 12-15 hrs (~80-85% reduction)**
- Leveraged **Sourcegraph Cody with Anthropic Claude (Opus/Sonnet)** to accelerate development and iteration of automation logic, reducing implementation time and improving reliability of complex data workflows
- Built **Python/Streamlit automation playbooks**: Finance Playbook (37-38 tasks), Customer Master Playbook (8-9 steps, ~1M+ B2B records), plus data extraction/validation workflows
- Developed **ImpEx generation engine** for automated HAC import/export scripts — B2B unit hierarchy, address transformations, and partner configurations
- Implemented **FlexibleSearch query automation** and **Groovy script execution** for complex data manipulations within SAP HAC
- Built **MongoDB integration layer** with PII data masking, chunked operations, and encrypted session management for secure multi-system authentication
- Created comprehensive **logging, error handling, and execution tracking** mechanisms ensuring audit compliance and eliminating manual sequencing errors

Data Analyst

Feb 2024 – Jun 2024

MassDOT (Massachusetts Department of Transportation)

Massachusetts, USA

- Designed and developed **automated data pipelines** using SQL and Python to extract, transform, and analyze CDLIS/SPEXS transaction data from the **ATLAS database**, reducing manual data processing cycles by **50%**
- Analyzed **AAMVA CD31 transaction response times** to measure Massachusetts' compliance with federal CDLIS timeliness standards, identifying bottleneck patterns in state-to-state data exchanges and delivering root-cause findings to OOS Unit leadership
- Performed systematic **CDLIS error pattern analysis** — categorizing error codes by frequency and severity, providing prioritized remediation recommendations that improved data accuracy
- Built **interactive Tableau dashboards** visualizing CDLIS transaction volumes, error rate trends, timeliness compliance metrics, and state comparison benchmarks, improving stakeholder decision-making efficiency by **35%**
- Supported **legacy-to-SPEXS database migration** by validating data integrity of migrated records, developing reconciliation scripts to identify conversion failures, and ensuring zero data loss during the ATLAS transition
- Developed **automated compliance reports** tracking Massachusetts' adherence to **FMCSA CDL requirements** and AAMVA data exchange standards, reducing manual reporting effort by **40%**
- Profiled ATLAS database records to assess **data quality metrics** — identifying duplicate CDL records, incomplete driver history entries, and inconsistent state pointer data, delivering a data quality scorecard to the migration team

Integration Consultant → Senior Data Engineer → Data Analyst

Sep 2016 – Jul 2021

Larsen & Toubro Infotech (LTI)

India & Johannesburg, South Africa

Integration Data Analyst

Johannesburg, South Africa — Client: Liberty Insurance (Point of Contact, 1.5 years)

- Reviewed Functional Requirement documents and built **analytical dashboards** boosting SLA turnaround time by **15%** and securing new business for LTI
- Facilitated **Scrum ceremonies** with client and cross-functional teams, tracking backlogs and sprint progress, advancing team efficiency by **85%**

- Constructed and revamped **30+ SQL Jobs and Stored Procedures** for production-grade automation
- Connected **SQL databases with RStudio** using knitr, DBI, and RSQLite libraries for data analysis and business intelligence delivery

Senior Integration Data Engineer

Johannesburg, South Africa — Client: Liberty Insurance

- Developed **data mapping and migration scripts** for **25+ applications**, enhancing client operations by **30%**
- Implemented **Azure DevOps CI/CD Pipeline** to restructure code repositories, increasing team efficiency by **75%**
- Led **BizTalk migration** (2010 → 2016), increasing system functionality and productivity by **50%**

Integration Consultant

India - Client: OKQ8

- Estimated migration scope for **850+ BizTalk applications**, proposing design optimizations reducing cost by **18%**
- Analyzed business requirements and data mapping for **70+ interfaces**, generating revenue exceeding **\$700,000**
- Automated **Incident Management** via ServiceNow and system health checks, enhancing support turnaround by **90%**
- Trained **8 cross-functional team members** and documented SOPs, improving support efficiency by **57%**

EDUCATION

Master of Science in Analytics

2023

Northeastern University, Boston, MA

Concentration: Statistical Modeling & Mathematics | GPA: 3.97/4.0

Bachelor of Technology, Electronics & Telecommunication Engineering

2016

Bharati Vidyapeeth University College of Engineering, India

KEY PROJECTS

- **AI Search Summarization Platform** — Production-ready generative AI search using RAG, LangChain, Claude LLMs, and AWS serverless architecture (Waters Corporation)
- **SAP Backoffice Automation Suite** — 45+ automated tasks with ImpEx generation, MongoDB integration, delivering 80-85% effort reduction (Waters Corporation)
- **KDI Store Segmentation** — Engineered customer segmentation strategy by revenue and product assortment; presented Tableau/QlikView dashboards to CEO & CTO (Northeastern)
- **Predictive Analytics Models** — Deployed Linear, Logistic, KNN, SVM, Decision Tree, Random Forest, Gradient Boost & Neural Network models for forecasting (Northeastern)

CERTIFICATIONS & ACHIEVEMENTS

- **Service Delivery Excellence Award** — Recognized for outstanding client interaction and relationship management at LTI
- **Graduate Teaching Assistant** — Predictive & Business Analytics at Northeastern University; conducted lab sessions in R and Python, tutored students, and supported curriculum development
- Learning Microsoft SQL Server 2019 | Querying Microsoft SQL Server 2019 | Data Analysis Industry Training — Brainnest | Agile Project Management
- Active competitor in organized cricket tournaments — **MiLC USA, MSCL USA, and LMS South Africa** — demonstrating cross-cultural teamwork and competitive discipline